

assignment-3

June 13, 2026

1. Create a class named Student with attributes:

- name
- age
- course
- Create an object and print all the details.

```
[8]: class Student:
      name = "Kartik Verma"
      age = 20
      course = "B.Tech Data Science"
st = Student()
print(st.name)
print(st.age)
print(st.course)
```

Kartik Verma

20

B.Tech Data Science

2. Create a class named Car with attributes:

- brand
- model
- price
- Create two objects and display their details.

```
[38]: class Car:
      brand = ""
      model = ""
      price = 0

car1 = Car()
car1.brand = "Bugatti"
car1.model = "Veyron"
car1.price = 100000000

car2 = Car()
car2.brand = "Rolls Royce"
car2.model = "Phantom"
```

```

car2.price = 150000000

print("Car 1 Details")
print("Brand:", car1.brand)
print("Model:", car1.model)
print("Price:", car1.price)

print("\nCar 2 Details")
print("Brand:", car2.brand)
print("Model:", car2.model)
print("Price:", car2.price)

```

Car 1 Details
Brand: Bugatti
Model: Veyron
Price: 100000000

Car 2 Details
Brand: Rolls Royce
Model: Phantom
Price: 150000000

3. Create a 'class' named Employee 'and' use a constructor '(init)' to initialize:

- employee_id
- name
- salary
- Print the employee details.

```

[14]: class Employee:
    def __init__(self, employee_id, name, salary):
        self.employee_id = employee_id
        self.name = name
        self.salary = salary

    def display(self):
        print("Employee ID:", self.employee_id)
        print("Name:", self.name)
        print("Salary:", self.salary)

emp = Employee(101, "Kartik Verma", 50000)
emp.display()

```

Employee ID: 101
Name: Kartik Verma
Salary: 50000

4. Create a class named Rectangle with attributes:

- length

- width
- Create a method to calculate and print the area of the rectangle.

```
[15]: class Rectangle:
    def __init__(self, length, width):
        self.length = length
        self.width = width

    def area(self):
        area = self.length * self.width
        print("Area of Rectangle:", area)

rect = Rectangle(10, 5)
rect.area()
```

Area of Rectangle: 50

5. Create a class named Circle with an attribute radius.
 - Create a method that calculates and prints the area of the circle.

```
[17]: class Circle:
    def __init__(self, radius):
        self.radius = radius

    def area(self):
        area = 3.14 * (self.radius**2)
        print("Area of Circle:", area)

c1 = Circle(15)
c1.area()
```

Area of Circle: 706.5

6. Create a class named BankAccount with:
 - account_holder
 - balance
 - Create methods:
 - deposit(amount)
 - withdraw(amount)
 - Display the updated balance after each transaction.

```
[21]: class BankAccount:
    def __init__(self, account_holder, balance):
        self.account_holder = account_holder
        self.balance = balance

    def deposit(self, amount):
        self.balance += amount
```

```

    print("Deposited:", amount)
    print("Updated Balance:", self.balance)

    def withdraw(self, amount):
        if amount <= self.balance:
            self.balance -= amount
            print("Withdrawn:", amount)
            print("Updated Balance:", self.balance)
        else:
            print("Insufficient Balance!")

account = BankAccount("Kartik Verma", 10000000)
account.deposit(50000000)
account.withdraw(30000000)

```

```

Deposited: 50000000
Updated Balance: 60000000
Withdrawn: 30000000
Updated Balance: 30000000

```

7. Create a class named Animal with a method sound().
 - Create a child class Dog that overrides the sound() method and prints “Bark”.

```

[23]: class Animal:
        def sound(self):
            print("Animal makes a sound")

        class Dog(Animal):
            def sound(self):
                super().sound()
                print("Bark")

dog = Dog()
dog.sound()

```

```

Animal makes a sound
Bark

```

8. Create a class named Person with attributes:
 - name
 - age
 - Create a child class Student with an additional attribute:
 - roll_number
 - Display all the details using an object of the Student class.

```

[26]: class Person:
        def __init__(self, name, age):

```

```

        self.name = name
        self.age = age

class Student(Person):
    def __init__(self, name, age, roll_number):
        super().__init__(name, age)
        self.roll_number = roll_number

    def display(self):
        print("Name:", self.name)
        print("Age:", self.age)
        print("Roll Number:", self.roll_number)

s1 = Student("Kartik Verma", 20, 101)
s1.display()

```

Name: Kartik Verma
Age: 20
Roll Number: 101

9. Create a class named Calculator with methods:

- add()
- subtract()
- multiply()
- divide()
- Take two numbers as input and perform all operations.

```

[31]: class Calculator:
    def add(self, a, b):
        print("Addition:", a + b)

    def subtract(self, a, b):
        print("Subtraction:", a - b)

    def multiply(self, a, b):
        print("Multiplication:", a * b)

    def divide(self, a, b):
        if b != 0:
            print("Division:", a / b)
        else:
            print("Division by zero is not allowed")

num1 = float(input("Enter first number: "))
num2 = float(input("Enter second number: "))

calc = Calculator()

```

```
calc.add(num1, num2)
calc.subtract(num1, num2)
calc.multiply(num1, num2)
calc.divide(num1, num2)
```

Addition: 54.0

Subtraction: 46.0

Multiplication: 200.0

Division: 12.5

10. Create a class named LibraryBook with attributes:

- book_name
- author
- price
- Create a method display_book_info() that prints all book details. Create at least three book objects and display their information.

```
[36]: class LibraryBook:
    def __init__(self, book_name, author, price):
        self.book_name = book_name
        self.author = author
        self.price = price

    def display_book_info(self):
        print("Book Name:", self.book_name)
        print("Author:", self.author)
        print("Price:", self.price)
        print("-----")

book1 = LibraryBook("Hii", "Kartik Verma", 999)
book2 = LibraryBook("Hello", "Anurag Sharma", 899)
book3 = LibraryBook("Bye", "Ankit Kumar", 799)
book1.display_book_info()
book2.display_book_info()
book3.display_book_info()
```

Book Name: Hii

Author: Kartik Verma

Price: 999

Book Name: Hello

Author: Anurag Sharma

Price: 899

Book Name: Bye

Author: Ankit Kumar

Price: 799
